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Advanced Math

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

A

A. -13

Choice A is correct. It's given that $gx = fx + 5$. Since $fx = 4x^2 + 64x + 262$, it follows that $fx + 5 = 4x + 5^2 + 64x + 5 + 262$. Expanding the quantity $x + 5^2$ in this equation yields $fx + 5 = 4x^2 + 10x + 25 + 64x + 5 + 262$. Distributing the 4 and the 64 yields $fx + 5 = 4x^2 + 40x + 100 + 64x + 320 + 262$. Combining like terms yields $fx + 5 = 4x^2 + 104x + 682$. Therefore, $gx = 4x^2 + 104x + 682$. For a quadratic function defined by an equation of the form $gx = ax - h^2 + k$, where a, h, and k are constants and a is positive, gx reaches its minimum, k, when the value of x is h. The equation $gx = 4x^2 + 104x + 682$ can be rewritten in this form by completing the square. This equation is equivalent to $gx = 4x^2 + 26 + 682$, or $gx = 4x^2 + 26x + 169 - 169 + 682$. This equation can be rewritten as $gx = 4x + 13^2 - 169 + 682$, or $gx = 4x + 13^2 - 4169 + 682$, which is equivalent to $gx = 4x + 13^2 + 6$. This equation is in the form $gx = ax - h^2 + k$, where $a = 4$, $h = -13$, and $k = 6$. Therefore, gx reaches its minimum when the value of x is -13.

Choice B is incorrect. This is the value of x for which fx, rather than gx, reaches its minimum.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of x for which $fx - 5$, rather than $fx + 5$, reaches its minimum.

Q2 C

C. 4

Choice C is correct. It's given that the function P models the population of the city t years after 2005. Since there are 12 months in a year, 18 months is equivalent to $\frac{18}{12}$ years. Therefore, the expression $\frac{18}{12}x$ can represent the number of years in x 18-month periods. Substituting $\frac{18}{12}x$ for t in the given equation yields $P\frac{18}{12}x = 2901.04^{\frac{4}{6}\frac{18}{12}x}$, which is equivalent to $P\frac{18}{12}x = 2901.04^{x}$. Therefore, for each 18-month period, the predicted population of the city is 1.04 times, or 104% of, the previous population. This means that the population is predicted to increase by 4% every 18 months.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. Each year, the predicted population of the city is 1.04 times the previous year's predicted population, which is not the same as an increase of 1.04%.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3 B

B. 7

Choice B is correct. It’s given that the situation can be modeled by the equation $y = -100x^2 + 1{,}400x + 7{,}200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. Since the coefficient of the x^2 term is negative, the graph of this equation in the xy-plane opens downward and reaches its maximum value at its vertex. If a quadratic equation in the form $y = ax^2 + bx + c$, where a, b, and c are constants, is graphed in the xy-plane, the x-coordinate of the vertex is equal to $-\frac{b}{2a}$. For the equation $y = -100x^2 + 1{,}400x + 7{,}200$, $a = -100$, $b = 1{,}400$, and $c = 7{,}200$. It follows that the x-coordinate of the vertex is $-\frac{1{,}400}{2(-100)}$, or 7. Therefore, if the given equation is graphed in the xy-plane, the maximum of the graph occurs at an x-value of 7.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

A

A. $fx = x + 44^2$

Choice A is correct. Let x represent the side length, in inches, of square P. It follows that the perimeter of square P is $4x$ inches. It's given that square Q has a perimeter that is 176 inches greater than the perimeter of square P. Thus, the perimeter of square Q is 176 inches greater than $4x$ inches, or $4x + 176$ inches. Since the perimeter of a square is 4 times the side length of the square, each side length of Q is $\frac{4x + 176}{4}$, or $x + 44$ inches. Since the area of a square is calculated by multiplying the length of two sides, the area of square Q is $x + 44x + 44$, or $x + 44^2$ square inches. It follows that function f is defined by $fx = x + 44^2$.

Choice B is incorrect. This function represents a square with side lengths $x + 176$ inches.

Choice C is incorrect. This function represents a square with side lengths $176x + 44$ inches.

Choice D is incorrect. This function represents a square with side lengths $176x + 176$ inches.

Q5 D

D. $h = -16t - 1.8^2 + 51.84$

Choice D is correct. An equation representing the height above ground h, in meters, of a softball t seconds after it is launched by a machine from ground level can be written in the form $h = -at - b^2 + c$, where a, b, and c are positive constants. In this equation, b represents the time, in seconds, at which the softball reaches its maximum height of c meters above the ground. It's given that this softball reaches a maximum height of 51.84 meters above the ground at 1.8 seconds; therefore, $b = 1.8$ and $c = 51.84$. Substituting 1.8 for b and 51.84 for c in the equation $h = -at - b^2 + c$ yields $h = -at - 1.8^2 + 51.84$. It's also given that this softball hits the ground at 3.6 seconds; therefore, $h = 0$ when $t = 3.6$. Substituting 0 for h and 3.6 for t in the equation $h = -at - 1.8^2 + 51.84$ yields $0 = -a3.6 - 1.8^2 + 51.84$, which is equivalent to $0 = -a1.8^2 + 51.84$, or $0 = -3.24a + 51.84$. Adding $3.24a$ to both sides of this equation yields $3.24a = 51.84$. Dividing both sides of this equation by 3.24 yields $a = 16$. Substituting 16 for a in the equation $h = -at - 1.8^2 + 51.84$ yields $h = -16t - 1.8^2 + 51.84$. Therefore, $h = -16t - 1.8^2 + 51.84$ represents the height above ground h, in meters, of this softball t seconds after it is launched.

Choice A is incorrect. This equation represents a situation where the maximum height is 3.6 meters above the ground at 0 seconds, not 51.84 meters above the ground at 1.8 seconds.

Choice B is incorrect. This equation represents a situation where the maximum height is 51.84 meters above the ground at 0 seconds, not 1.8 seconds.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6

D

D. $(x+1)(x-2)$

Choice D is correct. According to the table, when x is -1 or 2, $p(x)=0$. Therefore, two x-intercepts of the graph of p are $(-1,0)$ and $(2,0)$. Since $(-1,0)$ and $(2,0)$ are x-intercepts, it follows that $(x+1)$ and $(x-2)$ are factors of the polynomial equation. This is because when $x=-1$, the value of $x+1$ is 0. Similarly, when $x=2$, the value of $x-2$ is 0. Therefore, the product $(x+1)(x-2)$ is a factor of the polynomial function p.

Choice A is incorrect. The factor $x-3$ corresponds to an x-intercept of $(3,0)$, which isn't present in the table. Choice B is incorrect. The factor $x+3$ corresponds to an x-intercept of $(-3,0)$, which isn't present in the table. Choice C is incorrect. The factors $x-1$ and $x+2$ correspond to x-intercepts $(1,0)$ and $(-2,0)$, respectively, which aren't present in the table.

Q7

B

B. 45

Choice B is correct. It's given that $fx = x - 44x - 46$, which can be rewritten as $fx = x^2 - 90x + 2{,}024$. Since the coefficient of the x^2-term is positive, the graph of $y = fx$ in the xy-plane opens upward and reaches its minimum value at its vertex. For an equation in the form $fx = ax^2 + bx + c$, where a, b, and c are constants, the x-coordinate of the vertex is $-\frac{b}{2a}$. For the equation $fx = x^2 - 90x + 2{,}024$, $a = 1$, $b = -90$, and $c = 2{,}024$. It follows that the x-coordinate of the vertex is $-\frac{-90}{21}$, or 45. Therefore, fx reaches its minimum when the value of x is 45.

Choice A is incorrect. This is one of the x-coordinates of the x-intercepts of the graph of $y = fx$ in the xy-plane.

Choice C is incorrect. This is one of the x-coordinates of the x-intercepts of the graph of $y = fx$ in the xy-plane.

Choice D is incorrect. This is the y-coordinate of the vertex of the graph of $y = fx$ in the xy-plane.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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